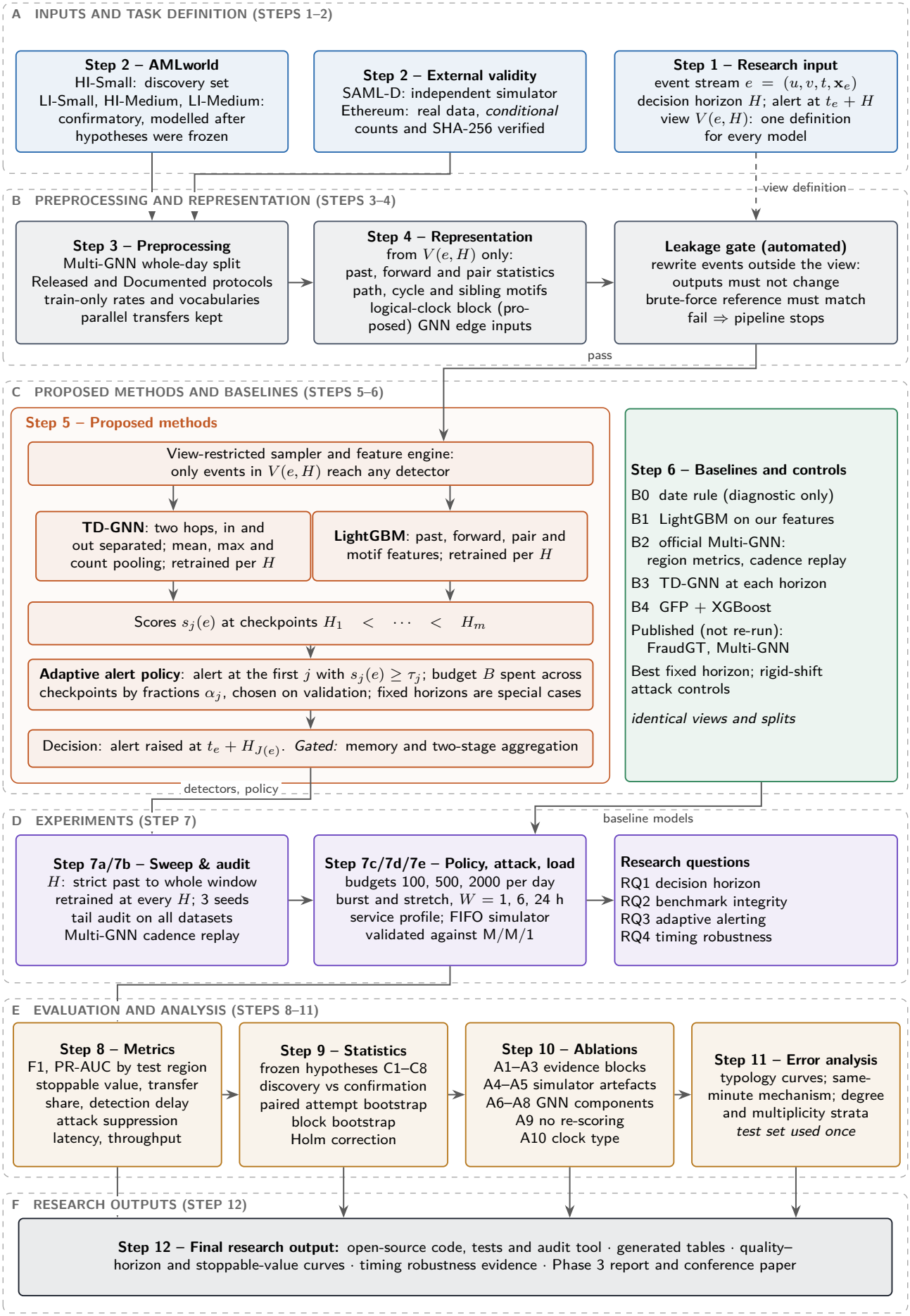

A INPUTS AND TASK DEFINITION (STEPS 1–2)
Step 2 – AMLworld
HI-Small: discovery set
LI-Small, HI-Medium, LI-Medium:
confirmatory, modelled after
hypotheses were frozen
Step 2 – External validity
SAML-D: independent simulator
Ethereum: real data, conditional
counts and SHA-256 verified
Step 1 – Research input
event stream $e = (u, v, t, \mathbf{x}_e)$
decision horizon H; alert at $t_e + H$
view $V(e, H)$: one definition
for every model
view definition
B PREPROCESSING AND REPRESENTATION (STEPS 3–4)
Step 3 – Preprocessing
Multi-GNN whole-day split
Released and Documented protocols
train-only rates and vocabularies
parallel transfers kept
Step 4 – Representation
from $V(e, H)$ only:
past, forward and pair statistics
path, cycle and sibling motifs
logical-clock block (pro-
posed) GNN edge inputs
Leakage gate (automated)
rewrite events outside the view:
outputs must not change
brute-force reference must match
fail ⇒ pipeline stops
pass
C PROPOSED METHODS AND BASELINES (STEPS 5–6)
Step 5 – Proposed methods
View-restricted sampler and feature engine:
only events in $V(e, H)$ reach any detector
TD-GNN: two hops, in and
out separated; mean, max and
count pooling; retrained per H
LightGBM: past, forward, pair and
motif features; retrained per H
Scores $s_j(e)$ at checkpoints $H_1 < \cdots < H_m$
Adaptive alert policy: alert at the first j with $s_j(e) \geq \tau_j$; budget B spent across
checkpoints by fractions α_j, chosen on validation; fixed horizons are special cases
Decision: alert raised at $t_e + H_{J(e)}$. Gated: memory and two-stage aggregation
Step 6 – Baselines and controls
B0 date rule (diagnostic only)
B1 LightGBM on our features
B2 official Multi-GNN:
region metrics, cadence replay
B3 TD-GNN at each horizon
B4 GFP + XGBoost
Published (not re-run):
FraudGT, Multi-GNN
Best fixed horizon; rigid-shift
attack controls
identical views and splits
detectors, policy
baseline models
D EXPERIMENTS (STEP 7)
Step 7a/7b – Sweep & audit
H: strict past to whole window
retrained at every H; 3 seeds
tail audit on all datasets
Multi-GNN cadence replay
Step 7c/7d/7e – Policy, attack, load
budgets 100, 500, 2000 per day
burst and stretch, $W = 1, 6, 24$ h
service profile; FIFO simulator
validated against M/M/1
Research questions
RQ1 decision horizon
RQ2 benchmark integrity
RQ3 adaptive alerting
RQ4 timing robustness
E EVALUATION AND ANALYSIS (STEPS 8–11)
Step 8 – Metrics
F1, PR-AUC by test region
stoppable value, transfer
share, detection delay
attack suppression
latency, throughput
Step 9 – Statistics
frozen hypotheses C1–C8
discovery vs confirmation
paired attempt bootstrap
block bootstrap
Holm correction
Step 10 – Ablations
A1–A3 evidence blocks
A4–A5 simulator artefacts
A6–A8 GNN components
A9 no re-scoring
A10 clock type
Step 11 – Error analysis
typology curves; same-
minute mechanism; degree
and multiplicity strata
test set used once
F RESEARCH OUTPUTS (STEP 12)
Step 12 – Final research output: open-source code, tests and audit tool · generated tables · quality–
horizon and stoppable-value curves · timing robustness evidence · Phase 3 report and conference paper